ST STITHIANS COLLEGE

Tour de Maths 2017

Leg 5

Question Paper MEMORANDUM

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5 Mark Questions

Question 1

Find the missing value.

1 + 4 $\rightarrow$ 5

2 + 5 $\rightarrow$ 12

3 + 6 $\rightarrow$ 21

8 + 11 $\rightarrow$ $\square$

MEMO

$$1\times4+1=5$$
$$2\times5+2=12$$
$$3\times6+3=21$$
$$8\times11+8=96$$

ANSWER: 96

Question 2

Find the missing power for base 2.

$$2^{2017}-2^{2016}=2^{\square}$$

MEMO

$$\begin{aligned}2^{2017}-2^{2016}&=2^{2016}(2-1)\\&=2^{2016}\end{aligned}$$

ANSWER: 2016

Question 3

It takes 10 minutes to fry one steak – 5 minutes from each side.

Two steaks can fit in your pan.

What is the shortest possible time to fry 3 steaks? Provide your answer in minutes.

MEMO

Steak 1	**Steak 2**	**Steak 3**	**Time**
side 1	side 1		5
side 2		side 1	5
	side 2	side 2	5
			15 MINS

ANSWER: 15

Question 4

How many ways are there to divide 3 identical sweets between 3 friends?

(Note: all sweets must be distributed, not everyone has to get the same amount of sweets, and people can receive 0 sweets)

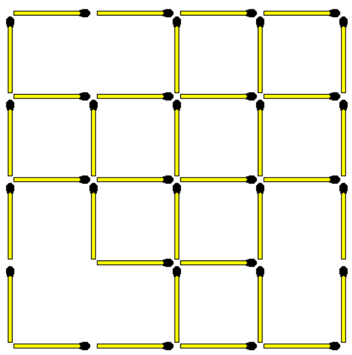
MEMO

	Person A	Person B	Person C
1	0	0	3
2	0	3	0
3	3	0	0
4	0	1	2
5	0	2	1
6	1	0	2
7	1	2	0
8	2	1	0
9	2	0	1
10	1	1	1

ANSWER: 10

Question 5

How many complete squares are in this matchstick puzzle?



MEMO

$(4\times4)=1$

$(3\times3)=1$

$(2\times2)=5$

$(1\times1)=9$

ANSWER: 16

Question 6

Jess, Nicole and Caitlyn are three sisters. The product of their ages is 175. Jess and Nicole are twins.

How old is Caitlyn?

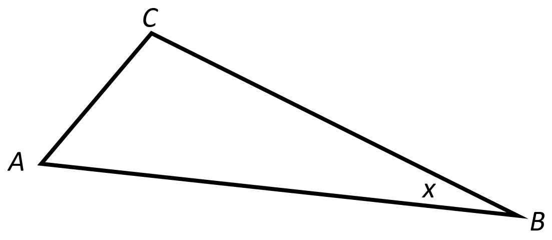
MEMO

$$Jess\times Nicole\times Caityn=175$$
$$x\times x\times y=175$$
$$x^2y=175$$
$$x^2y=5^2 7$$

ANSWER: 7

Question 7

In triangle ABC, the angle at A is three times greater than the angle at B and the angle at C is double the angle at A.
What is the value of the angle at A in degrees?



MEMO

$$3x+6x+x=180^\circ$$
$$10x=180^\circ$$
$$x=18^\circ$$
$$\hat{A}=54^\circ$$

ANSWER: 54

Question 8

$\sqrt{x+15}+\sqrt{x}=15$

Find the smallest integer value for x.

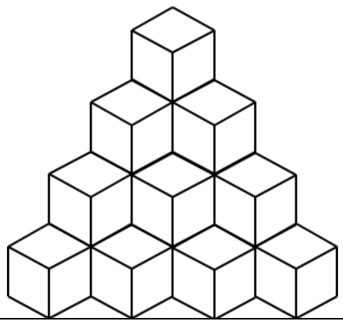
MEMO

$$\sqrt{x+15}+\sqrt{x}=15$$
$$\sqrt{x+15}=15-\sqrt{x}$$
$$x+15=225-30\sqrt{x}+x$$
$$30\sqrt{x}=210$$
$$\sqrt{x}=7$$
$$x=49$$

ANSWER: 49

Question 9

The following stack of blocks that is 4 cubes high is made up of 20 cubes.
How many cubes are required to build a similar stack that is 6 cubes high?



MEMO

ROW	BLOCKS	STRUCTURE
1	1	1
2	4	1+3
3	10	1+3+6
4	20	1+3+6+10
5	35	1+3+6+10+15
6	56	1+3+6+10+15+21

ANSWER: 56 cubes

Question 10

The product of two consecutive pages of a book is 11 556.

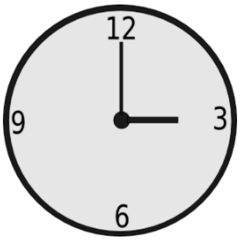
What is the page number of the page on the left?

MEMO

$$\sqrt{11556}=107.49$$

ANSWER: 107

10 Mark Questions

Question 11	MEMO
 Being an avid watch watcher, you discover that the hands of a watch are perpendicular at 3pm and at 9am. How many times a 24-hour day are the hands of a watch perpendicular to each other?	The minute hand moves 360 degrees in 60 minutes. This means that the angle of the minute hand is given by $6t$, where t is number of minutes past midnight. The hour hand moves 30 degrees in 60 minutes. This means that the angle of the hours hand is given by $\frac{1}{2}t$. The hands start together at midnight. The first time they make a 90 degree angle is when the minute hand has moved 90 degrees further than the hour hand, so this is given by the equation: $$6t = \frac{1}{2}t + 90$$ $$\frac{11}{2}t = 90$$ $$t = \frac{180}{11}$$ $$t \approx 16\,\text{min}$$ In other words about 16 minutes past midnight. The next time is when the minutes hand has gained another 180 degrees on the hour hand, and is 90 degrees behind it: $$6t = \frac{1}{2}t + 270$$ $$\frac{11}{2}t = 270$$ $$t = \frac{540}{11}$$ $$t \approx 49\,\text{min}$$ At about 11 minutes to 1 o'clock. For every 180 degrees that the minute hand gains on the hour hand there will be one 90 degree angle, so every $\frac{540}{11} - \frac{180}{11} = \frac{360}{11}\,\text{min}$ 24 hours is 1440 minutes. $1440 \div \frac{360}{11} = 44$ So every 24 hours there are 44 right angles between minute hand and second hand.

	ANSWER: 44

Question 12

1

3 5

7 9 11

13 15 17 19

21 23 25 27 29

Find the sum of the 30th row.

MEMO

ROW	SUM
1	1
2	8
3	27
4	64
5	125
n	n^3
30	27000

ANSWER: 27000

Question 13

This question paper consists of 20 questions: 10 questions for 5 marks, 5 questions for 10 marks and 5 questions for 20 marks each. A team of 4 members, Brad, Muhammad, Conner and Ben get all the 5 mark questions correct and 4 of the 10 mark questions correct.

What is the minimum amount of 20 mark questions they must answer correctly to score over 80% for the paper as a whole?

MEMO

VALUE	points achieved	possible
5 x 10	50	50
10 x 5	40	50
20 x 5	??	100
need	160	200

160-90=70 points but need to form in 20 point questions therefore need minimum of 4 of the 20 mark questions correct to score over 80%

ANSWER: 4

Question 14

Determine the value of:

$$100^2-99^2+98^2-97^2+\ldots-3^2+2^2-1^2$$

MEMO

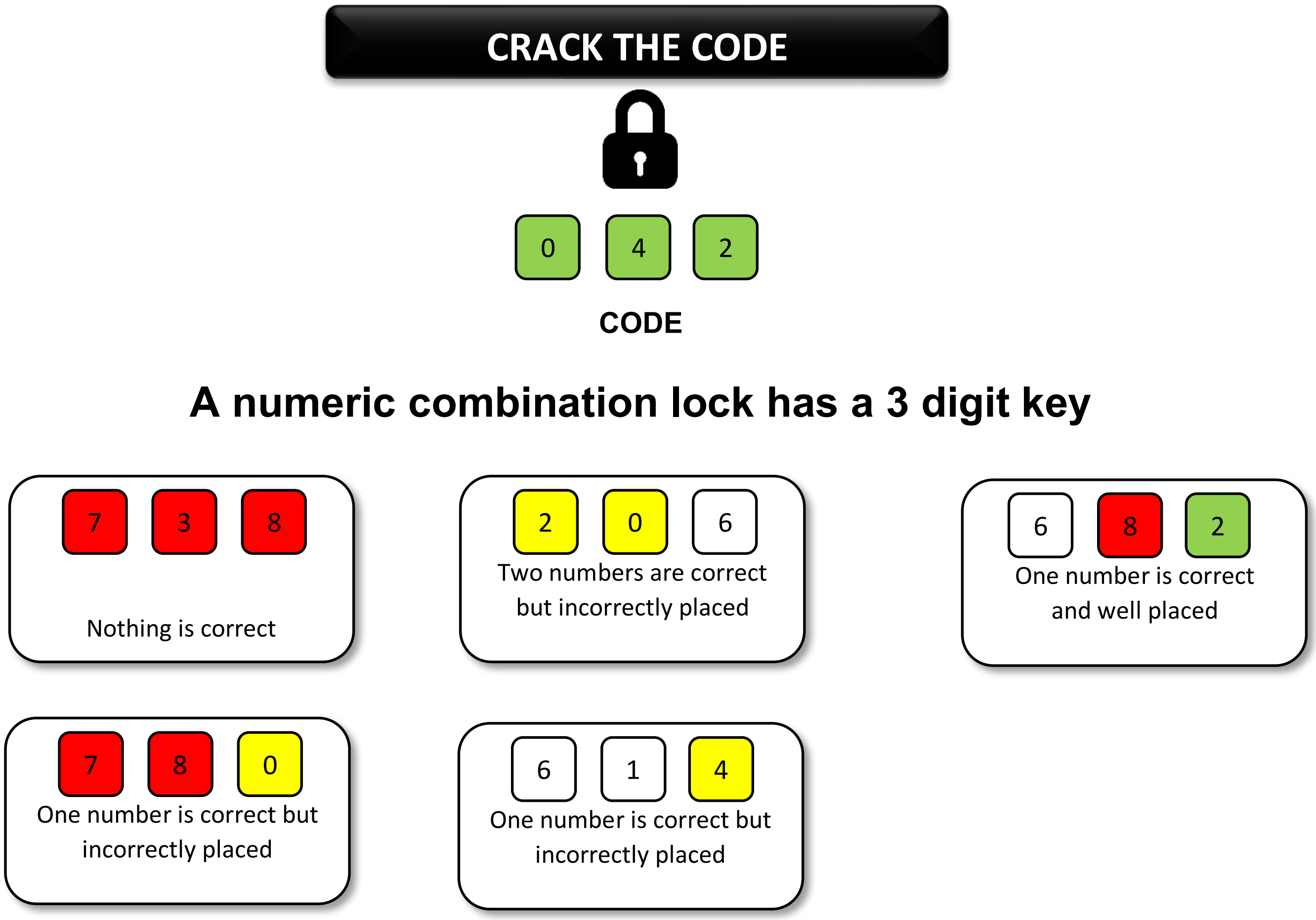
$$100^2-99^2+98^2-97^2+\ldots-3^2+2^2-1^2$$
$$=(100+99)(100-99)+(98+97)(98-97)+\ldots+(2+1)(2-1)$$
$$=100+99+98+97+\ldots+3+2+1$$
$$=(100+1)+(99+2)+(98+3)+\ldots+(51+50)$$
$$=101(50)$$
$$=5050$$

ANSWER: 5050

Question 15	**MEMO**
If x denotes any negative integer number, which of the following expressions will always have the greatest value? (A) $\sqrt{-x} \times \sqrt{-x}$ (B) $-(x^2+81)$ (C) $\left(\sqrt{x}\right)^8$ (D) $\sqrt{x^4}+1$ (E) $\left(\sqrt[3]{x^9}\right)^3$	(A) $\sqrt{-x} \times \sqrt{-x}$ $=\left(\sqrt{-x}\right)^2$ $=-x$ $=-(-\text{int})$ $=+\text{int}$ (B) $-(x^2+81)$ $=-((-\text{int})^2+81)$ $=-(\text{int}^2+81)$ $=-\text{int}^2-81$ (C) $\left(\sqrt{x}\right)^8$ $non-real$ (D) $\sqrt{x^4}+1$ $=x^2+1$ $=(-\text{int})^2+1$ $=+\text{int}^2+1$ (E) $\left(\sqrt[3]{x^9}\right)^3$ $=(x^{\frac{9}{3}})^3$ $=x^9$ $=(-\text{int})^9$ $=-\text{int}^9$ **ANSWER: D**

20 Mark Questions

Question 16



ANSWER: 042

Question 17

There are four buttons in a row as shown below. Two of them show happy faces, and two of them show sad faces. If we press a face, its expression turns to the opposite (e.g. a happy face turns into a sad face after the touch). In addition to this, the adjacent buttons also change their expressions.

What is the least number of times you need to press a button in order to get all happy faces?

ANSWER: 3

Question 18

$$\begin{array}{ccc} & A & B \\ + & C & D \\ \hline & E & F \end{array}$$

In this cryptogram, each letters represent a distinct (different) digit.
What is the minimum possible value of the 2-digit integer EF?

MEMO

ASSUMING ZERO CAN BE USED AS A TEN

	0	7
+	1	6
	2	3

ANSWER: 23

ASSUMING ZERO CAN'T BE USED AS A TEN

	2	5
+	1	4
	3	9

ANSWER: 39

Question 19

We define a relatively prime date to be a day for which the number of the month and the number of the day have no common factors other than 1. For example 22/5 (22 May) is such a day because 22 and 5 have no common factors other than 1. Determine the month with the smallest number of relatively prime days.

MEMO

1	2	3	4 1;2;4	5	6 1;2;3;6	7	8 1;2;4;8	9 1;3;9	10 1;2;5;10	11	12 1;2;3;4;6;12
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9
10	10	10	10	10	10	10	10	10	10	10	10
11	11	11	11	11	11	11	11	11	11	11	11
12	12	12	12	12	12	12	12	12	12	12	12
13	13	13	13	13	13	13	13	13	13	13	13
14	14	14	14	14	14	14	14	14	14	14	14
15	15	15	15	15	15	15	15	15	15	15	15
16	16	16	16	16	16	16	16	16	16	16	16
17	17	17	17	17	17	17	17	17	17	17	17
18	18	18	18	18	18	18	18	18	18	18	18
19	19	19	19	19	19	19	19	19	19	19	19
20	20	20	20	20	20	20	20	20	20	20	20
21	21	21	21	21	21	21	21	21	21	21	21
22	22	22	22	22	22	22	22	22	22	22	22
23	23	23	23	23	23	23	23	23	23	23	23
24	24	24	24	24	24	24	24	24	24	24	24
25	25	25	25	25	25	25	25	25	25	25	25
26	26	26	26	26	26	26	26	26	26	26	26
27	27	27	27	27	27	27	27	27	27	27	27
28	28	28	28	28	28	28	28	28	28	28	28
29		29	29	29	29	29	29	29	29	29	29
30		30	30	30	30	30	30	30	30	30	30
31		31		31		31	31		31		31
30	13	20	14	24	9	26	15	19	12	27	10

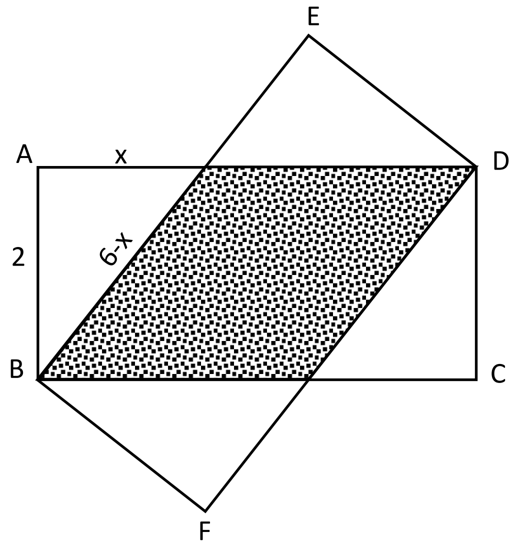
ANSWER: JUNE

Question 20

In the diagram below, two identical rectangles have been placed on top of each other.

AB = BF = 2 and AD = BE = 6

Find the area of the shaded region.



calculate x	**area of 1 rectangle**	**Area of shaded**
$(6-x)^2 = x^2+2^2$ $36-12x+x^2 = x^2+4$ $32=12x$ $x=\frac{32}{12}$ $x=\frac{8}{3}$	$area = 2\times 6$ $=12 units^2$	$shaded = area_1 - area(2\times\triangle)$ $=12-2(\frac{1}{2}\times\frac{8}{3}\times 2)$ $=12-\frac{16}{3}$ $=\frac{36-16}{3}$ $=\frac{20}{3}$ $=6,\dot{6} units^2$

ANSWER: $\frac{20}{3}$ **OR 6.6**